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SHORT REPORT

*Control of Movement*

# Characterization of limb representation in the pig's motor cortex

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## Abstract

Due to its large gyrencephalic brain, the pig is increasingly used for neuroscience research, especially for the preclinical testing of novel neuroprostheses. However, our understanding of the pig's motor system remains limited compared with the common species used for neuroscience research. Here, we aimed to characterize the forelimb and hindlimb representation of the pig motor cortex using intracortical microstimulation (ICMS). Three domestic pigs (*Sus scrofa*) were placed in a modified stereotactic frame and maintained under intravenous propofol sedation. We mapped the motor cortex using ICMS, applied at varying cortical coordinates and depths. For each site, we recorded the electrode depth eliciting the maximal limb response and determined the motor threshold. Responses were assessed visually and via electromyographic recordings. ICMS uncovered a large forelimb representation, with stereotypical contralateral responses. Conversely, the hindlimb representation was smaller and located within the interhemispheric fissure. The mean threshold of the five most responsive forelimb sites was $75 \pm 25$ μA, compared with $280 \pm 45$ μA for hindlimb sites ($P < 0.01$). A summation of stimulations in the hindlimb representation of the motor cortex unilaterally triggered bilateral alternating hindlimb movements. These results suggest that while the porcine cortex can directly command forelimb movements via the corticospinal pathway, cortical control of hindlimb likely relies on polysynaptic pathways through the brainstem, such as the corticoreticulospinal pathway.

**NEW & NOTEWORTHY** We identified a limited hindlimb representation on the medial surface of the pig motor cortex that, when repeatedly stimulated, produced bilateral alternating hindlimb movements. This suggests that the cortical control of hindlimb likely relies on polysynaptic pathways through the brainstem, such as the corticoreticulospinal pathway.

*intracortical microstimulation; motor cortex; neuroprosthesis; pig*

## INTRODUCTION

Rodents and nonhuman primates have long been the predominant animal models in neuroscience. The pig (*Sus scrofa*) is increasingly recognized as a valuable preclinical model for neuroscience research and neuroprosthesis biocompatibility testing (1, 2). The gyrencephalic pig brain shares closer anatomical (3, 4) and developmental (5) similarities with the human brain than commonly used small laboratory animals and has long been used in other medical fields, such as cardiology and transplantation, due to their similarity to humans in anatomical size and structure, physiology, and immunology (6).

To date, pigs have been used in preclinical studies for various neuroprosthetic devices, including deep brain stimulation (DBS) (7–9), vagus nerve stimulation (10), epidural spinal cord stimulation (11, 12), intradural spinal cord stimulation (13–18), and chronic cortical recordings with electrocorticographic (19) and micro-electrode arrays (20). Notably, pigs serve as the main preclinical model for biocompatibility testing of novel brain-machine interfaces, used by companies like Neuralink (Austin) (21) and Precision Medicine (New York) (22). Porcine models have also been developed for a range of nervous system disorders, including stroke (23, 24) and spinal cord injury (SCI) (25–30).

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Building on our previous work in rodent and feline models, where we developed a neuroprosthesis delivering intracortical microstimulation (ICMS) in synchrony with locomotion to modulate movements and improve motor recovery after SCI (31–35), we aimed to test the feasibility of studying cortical neuroprostheses on that model. Although the functional anatomy of the porcine spinal cord and nerve roots is well characterized (36, 37), the cortical representation of the pig hindlimb remains poorly defined. Transcranial electrical stimulation can generate hindlimb movements (38, 39), but less robustly than forelimb responses. The only prior ICMS studies in pigs date back to 1966 (40) and 1986 (41), with minimal description of the hindlimb representation. Moreover, tracing studies suggest that the porcine corticospinal tract does not extend beyond the cervico-thoracic levels (42–44), implying that hindlimb responses to cortical stimulation may rely predominantly on polysynaptic pathways.

In this study, we used ICMS to derive cortical motor maps in three anesthetized pigs, enabling a detailed characterization of both the hindlimb and forelimb motor representations.

## MATERIALS AND METHODS

### Animals

All procedures adhered to the Canadian Council on Animal Care guidelines and were approved by the Comité d'Éthique et de l'Expérimentation Animale (CEEA; animal ethics committee) at Hôpital du Sacré-Coeur de Montréal. Experiments were conducted on domestic pigs (*Sus scrofa*) purchased from *Ferme Gérard Riopel*. Three female pigs (2–2.5 mo old, weighing 21, 23, and 24 kg) were used. Animals were housed individually with constant access to water and were fed three times daily.

### Intracortical Microstimulation

Animals were preanesthetized with intramuscular injections of ketamine (20 mg/kg), xylazine (2 mg/kg), buprenorphine (0.05 mg/kg), acepromazine (0.05 mg/kg), and atropine (0.05 mg/kg). After intubation, pigs were placed under assisted ventilation and maintained on a heating mat to maintain body temperature above 35°C. The head was secured in a modified stereotactic frame using external auditory canal bars and a hard palate plate. The torso was elevated to allow unrestricted motion of the forelimbs and hindlimbs. Anesthesia was maintained with intravenous propofol (10 mg/mL, 10–30 mL/h), adjusted to preserve evoked ICMS responses. Optimal anesthesia was characterized by mild spontaneous limb twitching; lighter anesthesia (tachycardia or increased twitching) was managed with propofol boluses. Electrocardiogram (EKG) monitoring was continuous. The experimental setting is summarized in Fig. 1*A*. A large craniotomy was performed bilaterally to expose the motor cortex and surrounding gyri, spanning 2-cm posterior and 3-cm anterior to the coronal suture, and ~3 cm lateral to the sagittal suture. The dura was opened in a "C" shape and reflected toward the midline to preserve the superior sagittal sinus. If necessary, cortical veins adhering to the dura were coagulated and cut to facilitate proper reflection of the dura. The cortical surface was irrigated with saline heated to body temperature. The motor cortex was localized bilaterally between the sulcus ansatus posterolaterally and the sulcus cruciatus anteromedially (3, 40) (Fig. 1*B*). A micromanipulator was used to determine the coordinates of the motor cortex boundaries, and a grid of 1-mm squares was overlaid to guide electrode penetrations. Glass-coated platinum/iridium microelectrodes with an input impedance of 0.5 ± 0.1 MΩ (1,000 Hz, 10 nA) were used (FHC). Electrodes were advanced into the neocortex to a depth of 0.5 mm using a microdrive (David Kopf Instruments model 2650), with 0.5-mm increments until the area of peak responses was determined. An isolated pulse stimulator was used to deliver electrical current (Model 2100, A-M Systems). A ground stimulation lead was placed in contact with the neck muscles. Stimulation trains consisted of 13 biphasic cathodal-lead pulses, each with a 200-μs duration at a frequency of 333 Hz, with a train rate of 1 Hz. Current intensity was gradually increased from 0 μA to 1 mA maximum. Threshold was defined as the lowest intensity consistently producing a movement. Sites unresponsive at 1 mA were considered nonresponsive. We sampled each coordinate at multiple depths to identify the depths evoking motor responses at the lowest amplitude. Throughout the surgery, anesthetic levels were monitored by verifying evoked movements at previously defined positive-response sites and assessing the borders of nonresponsive points. To assess motor responses, we used electromyography (tibialis anterior and vastus lateralis bilaterally) and high frame-rate video recording (120 Hz, Teledyne FLIR) for visual analysis of evoked movements. A light-emitting diode synchronized with stimulator output was fixed to the stereotaxic frame in the camera field-of-view. Motor maps were generated from positive-response sites, with each site representing 1 $mm^2$. The hindlimb representation was the primary focus; forelimb mapping was partial, and facial (corticobulbar) representation was only broadly outlined. We performed ICMS bilaterally in three pigs in total, systematically applying penetrations 1 mm apart across the motor cortex. At each site, stimulations were delivered at depths from 1 to 7 mm in 1-mm increments, resulting in ~700 stimulation sites. At the end of the experiment, animals were deeply anesthetized with isoflurane and euthanized with intravenous KCl. Brains were extracted, postfixed with 16% paraformaldehyde (PFA) solution for 1 mo, then cryoprotected in a solution of 30% sucrose in 0.1 M PBS for 1 wk. The brain was frozen, sectioned around the motor cortex, and cut into 40-μm thick sagittal sections (3 hemispheres) and coronal sections (3 hemispheres) using a cryostat. The sections were stained with Luxol Fast Blue (0.1%, Sigma, S3382) and Cresyl Violet (0.5%, Alfa Aesar, J64318). Bright-field microscopy images were taken at ×4 magnification (Olympus BX63).

### Statistical Analyses

All data are displayed as means ± SE. We assessed the normal distribution of results with the Kolmogorov–Smirnov test. Statistical evaluations were performed using two-sided Student's *t* test or Mann–Whitney *U* test based on data normality. We performed analyses were performed to assess



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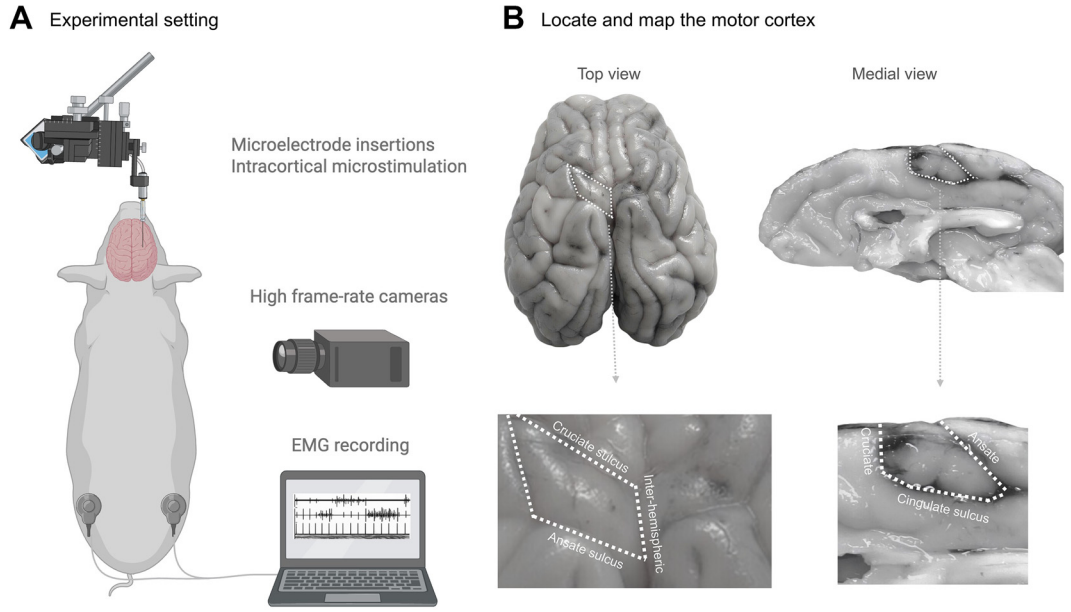



**Figure 1.** Experimental setup for intracortical microstimulation. *A*: a modified stereotactic frame and micromanipulator were used to insert a microelectrode at precise coordinates within the motor cortex. Responses to intracortical microstimulation were measured with high frame-rate video recording and electromyography (EMG). *B*: visual representation of the pig's motor cortex, from a superior view (*left*) and medial view (*right*). Below, the relevant anatomy is highlighted, notably the cruciate sulcus, ansate sulcus, cingulate sulcus, and interhemispheric fissure. Panel 1*A* was created with a licensed version of BioRender.com.

group differences in the minimum current intensity required to elicit forelimb and hindlimb movements during ICMS (five best sites; all sites). We applied a threshold for statistical significance at $P = 0.05$.

## RESULTS

### Intracortical Microstimulation

By applying ICMS bilaterally in three pigs (more than 700 stimulation sites), we determined the minimal current required to evoke a visible limb movement, identified the depth of maximal response, and derived a somatotopic motor map (Fig. 2, Supplemental Video S1). The sites evoking maximal hindlimb motor responses were at the medial edge of the motor cortex, whereas sites evoking a predominant forelimb response occupied a larger area of the grid centrally and laterally (Fig. 2, *A* and *B*); stimulation of the lateral and superior aspect of forelimb motor map evoked various facial responses, including the lips, tongue, and neck (Fig. 2, *A* and *B*). As we focused on cortical limb representation, we did not specifically map corticobulbar outputs in this study. The histological analysis of electrode tracts allowed to locate the layer V of the motor cortex at around 2 mm from the cortical surface (Fig. 2*C*). Electrode penetrations can cause a slight depression of the cortex, leading to a mild overestimation of penetration depths. As such, responses obtained between 2 and 4 mm of depths were considered cortical responses and electrode penetrations at depths of 5–7 mm were considered as white-matter responses (within the corona radiata) (Fig. 2*A*). At some coordinates, forelimb movements were often maximally evoked superficially (2–3 mm depths, corresponding to layer V), whereas hindlimb movements were elicited deeper down (6–7 mm depths, presumably stimulating white-matter fibers from the medial aspect of the cortex joining the corona radiata; Fig. 2, *B* and *C*). Taking these findings into consideration, we adjusted our approach for the ICMS mapping performed on the third pig, limiting penetration depth to 4 mm, except in our most medial penetrations adjacent to the interhemispheric fissure (to more precisely map the medial cortical surface). Electrode penetrations at the medial margin of the motor cortex (along the interhemispheric fissure) were used to map the medial aspect of the motor cortex (Fig. 2*D*, unfolded map of the pig's motor cortex).

### Hindlimb Movements Generated by ICMS

Hindlimb responses were generally subtler than forelimb responses when using the same stimulation amplitude. At forelimb sites, robust movements were obtained with currents as low as 50–100 µA, whereas hindlimb movements typically required >300 µA (Fig. 2*B*). The mean amplitude threshold of the five most responsive forelimb sites was 75 ± 25 µA, compared with 280 ± 45 µA for the hindlimb. Overall recorded forelimb and hindlimb sites, amplitude thresholds were lower for the forelimb (197 ± 115 µA) than the hindlimb (440 ± 171 µA; $P < 0.001$). Hindlimb responses often required repeated stimulations and were less consistent across trials, in contrast to the reproducible contralateral forelimb responses. Lateralization of responses also differed between forelimb and hindlimb: forelimb responses were strictly contralateral, whereas hindlimb responses were frequently ipsilateral or bilateral. Most hindlimb responses consisted of small twitches along the entire leg, often with a subtle flexion of the contralateral hooves or fetlock (Supplemental Video S1). ICMS applied to the medial portion of the motor cortex often triggered shoulder movements that frequently coincided with hindlimb movements. In contrast,



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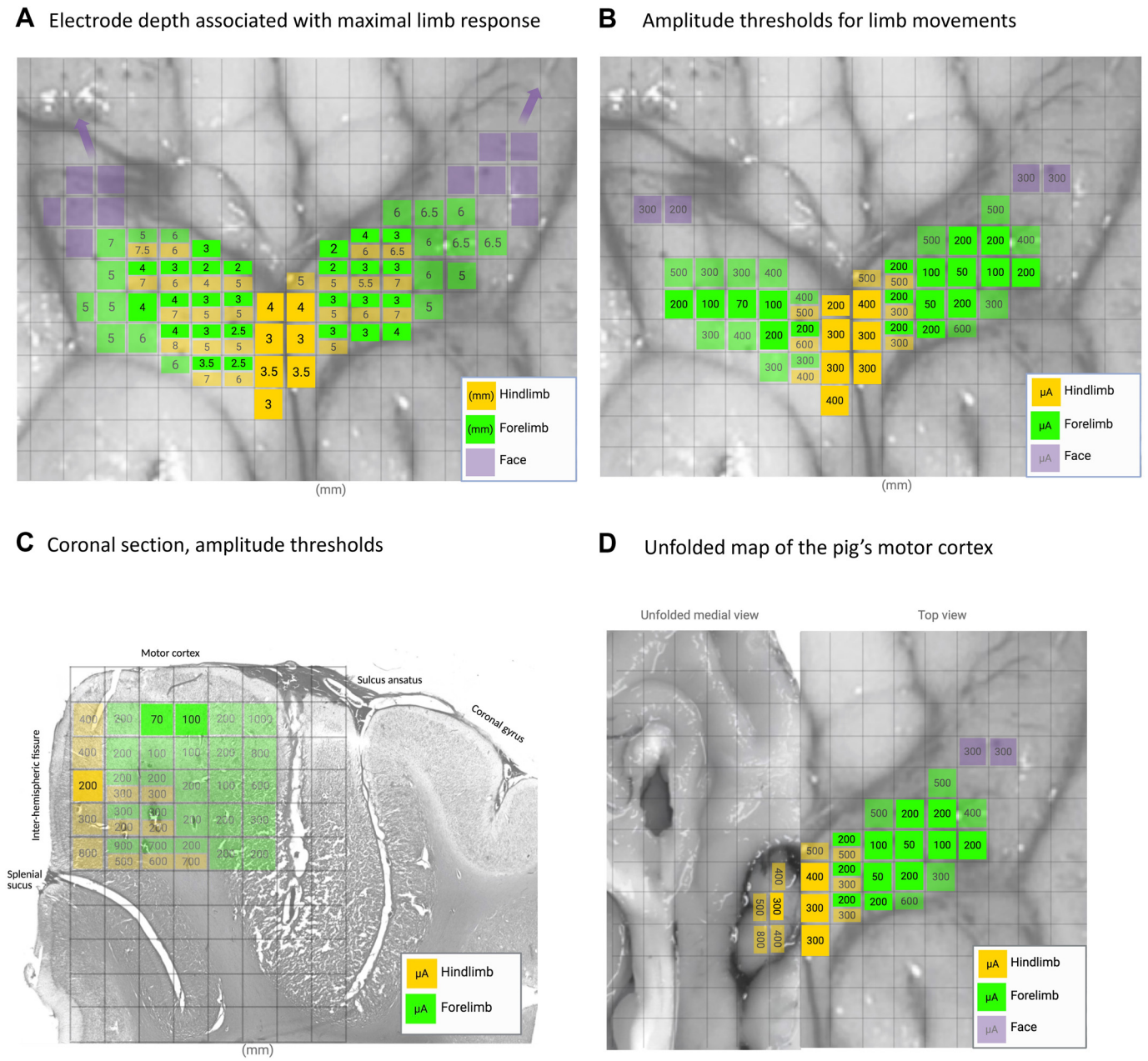



**Figure 2.** Cortical and subcortical map of the pig motor cortex. *A*: summary of cortical and subcortical mapping; electrode depths associated with maximal hindlimb (HL) responses, indicated in mm in yellow boxes. Sites for which the depth with the maximal responses was 5 mm or more (within subcortical white matter) are shaded. We roughly characterized the facial representation at the border of the forelimb representation; however, we did not attempt to fully characterize this facial (corticobulbar) representation in terms of motor map borders, movement types (snout, neck, upper and lower lips, tongue), or amplitude thresholds. *B*: summary of the cortical mapping. Amplitude thresholds (μA) of intracortical microstimulation to evoke forelimb and hindlimb movements are indicated. Only responses at penetrations of 2–4 mm are represented. Regions with higher amplitude thresholds are shaded. *C*: coronal section of the pig's motor cortex, highlighting the transition of forelimb responses superficially and hindlimb responses deeper into the white matter. *D*: unfolded map of the pig's motor cortex. The *right* shows the surface mapping of the right hemisphere. On the *left*, the medial view of the pig's cortex was rotated and adjusted for the cruciate and ansate sulci to align.

lateral stimulations resulted in clear flexion of the elbow and carpal joint (Supplemental Video S1).

### Bilateral Alternating Responses of Hindlimbs to Unilateral Cortical Stimulation

Stimulation of the hindlimb representation frequently produced bilateral hindlimb movements (Supplemental Video S2, Fig. 3, *A* and *B*). Although forelimb responses were consistently contralateral, hindlimb responses included ipsilateral or bilateral movements. In *pig 3*, across 189 stimulations of the hindlimb representation, 48% triggered contralateral, 38% ipsilateral, and 14% bilateral hindlimb movements. Response patterns depended on stimulation frequency. With long interstimulation intervals (>2 s), responses were predominantly contralateral (90%, 18/20). At shorter intervals (<0.8 s), ipsilateral (51%, 49/96) and



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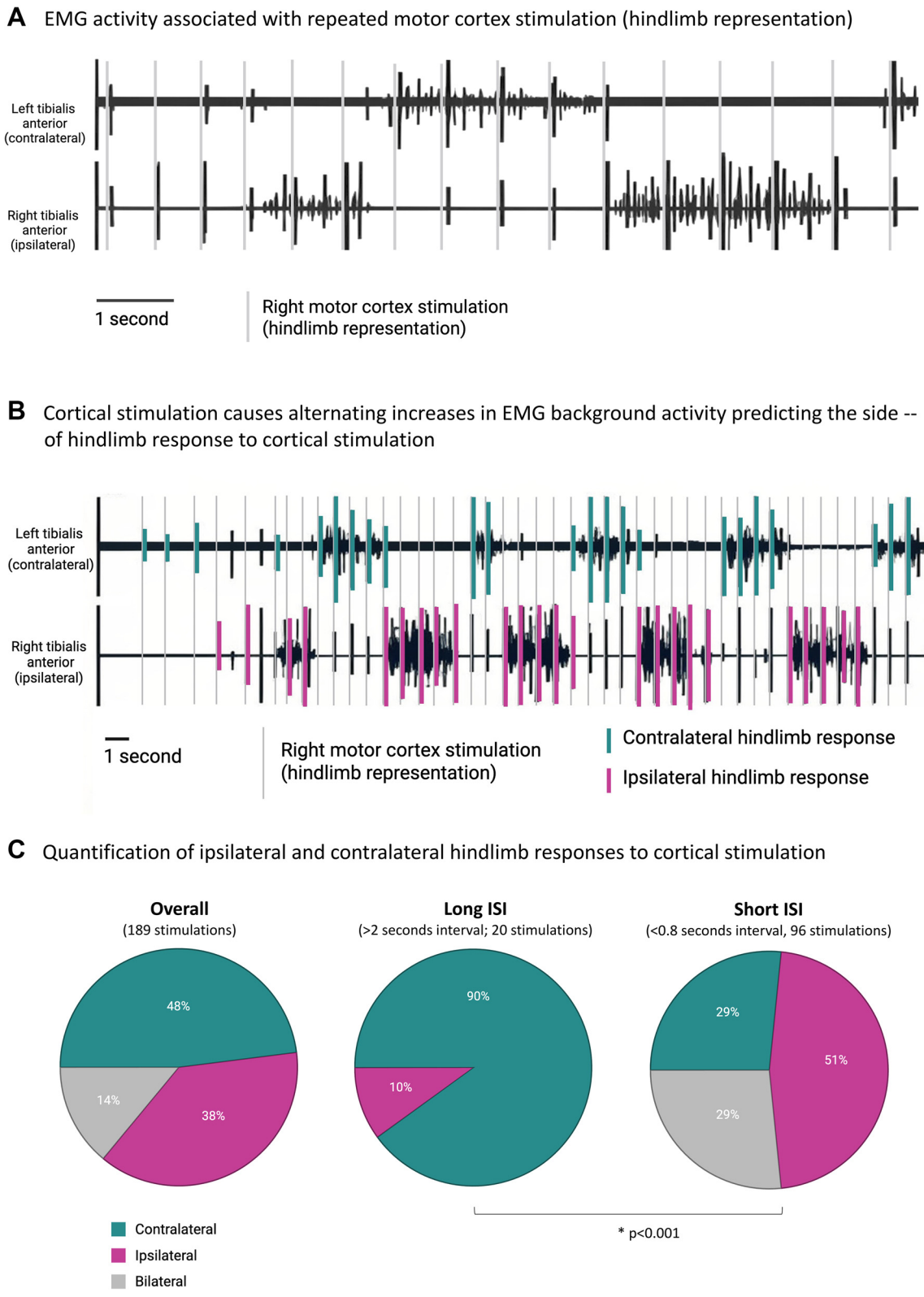



**Figure 3.** Repeated stimulation of the hindlimb representation of the pig's motor cortex causes alternating increases in background activity, associated with alternating bilateral responses to stimulation. *A*: zoomed-in EMG trace showing alternating bursts of tibialis anterior background activity during repeated stimulation of the leg representation of the pig's motor cortex. Cortical stimulation is represented with thin gray lines. *B*: cortical stimulation causes alternating increases in EMG background activity, predicting the side of hindlimb response to cortical stimulation. Thin gray lines represent stimulation pulses, while magenta and cyan lines represent evoked responses of the hindlimbs (visual responses). *C*: quantification of ipsilateral and contralateral hindlimb responses to cortical stimulation. ISI, interstimulus interval.



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bilateral (29%, 28/96) movements increased, while contralateral movements decreased to 29% (28/96) (Fig. 3*C*). At higher stimulation frequencies (>1 Hz), cycles of 2–4 responses typically alternated between hindlimbs (Fig. 3*B*). Interestingly, in *pig 3*, these alternating hindlimb responses were abolished by ketamine injection (10 mg/kg), with preservation of forelimb responses.

EMG analysis revealed alternating bursts of background activity consisting of short (40 ms) discharges at 10–16 Hz. Bursts lasted 0.5–4.5 s and alternated irregularly, occurring in cycles ranging from 1.3 s to 11 s. Interestingly, the dominant motor response (whether ipsilateral or contralateral movement) coincided with the leg exhibiting a background burst at the time of cortical stimulation. This pattern suggests that the background activity of the muscles in a given limb may influence the direction of movement evoked by cortical stimulation (Fig. 3*B*). In other words, we observed two distinct phenomena: *1*) the induction of alternating EMG bursts, which occur at a much lower frequency than the stimulus train and are not phase-locked to it, and *2*) a dependence of the amplitude of the rapid, stimulus-locked response on alternation phase, consistent with an "automatic gain scaling" effect (45). The duration and periodicity of the alternating bursts were highly variable, with shorter bursts (0.5 s) alternating on cycles of around 1.3 s, while longer bursts (4.5 s) alternated on cycles of ~11 s (Fig. 4). Finally, we quantified the amplitude of the EMG response to 100 consecutive cortical stimulations, depending on the underlying state of background activity (high background activity, low background activity). We normalized the data as a fraction of the maximal amplitude observed on the channel (0–100%). When the cortical stimulation occurred during the burst of hindlimb EMG activity, the amplitude of the EMG response was almost invariably maximal (mean 90% of maximal amplitude), even in the EMG ipsilateral to the cortical stimulation. When the cortical stimulation occurred during low background activity, the response was much lower (mean 29% of maximal amplitude; $P < 0.0001$), and almost half of stimulations did not trigger any EMG response.

## DISCUSSION

In this study, we characterized the forelimb and hindlimb representations of the pig motor cortex. We found that the hindlimb motor representation is relatively small compared with the forelimb, and that movements evoked by cortical stimulation were more consistent and pronounced in the face and forelimb than in the hindlimbs. Our results align with previous ICMS studies in pigs. In Breazile et al. (40), a summation of 3–4 monophasic square-wave stimuli of 4 ms, 7–10 V in the medial-caudal region of the motor cortex could elicit isolated hindlimb movements, a region corresponding to the primary somatosensory hindlimb representation, which is on the medial surface of the cruciate gyrus, buried within the interhemispheric fissure (46). The discrepancy between the forelimb and hindlimb representation in the motor cortex can be explained by the anatomy of the corticospinal tract in pigs. Complete resection of the pig's motor cortex does not significantly impair locomotion (41), and large cortical and subcortical strokes (middle cerebral artery occlusion) only produce subtle impairments in gait and forelimb movements during locomotion (23). In various tracing studies, the corticospinal tract was shown to reach the cervical level, but never below the thoracic level (42, 44). Comparable findings have been reported in sheep, where

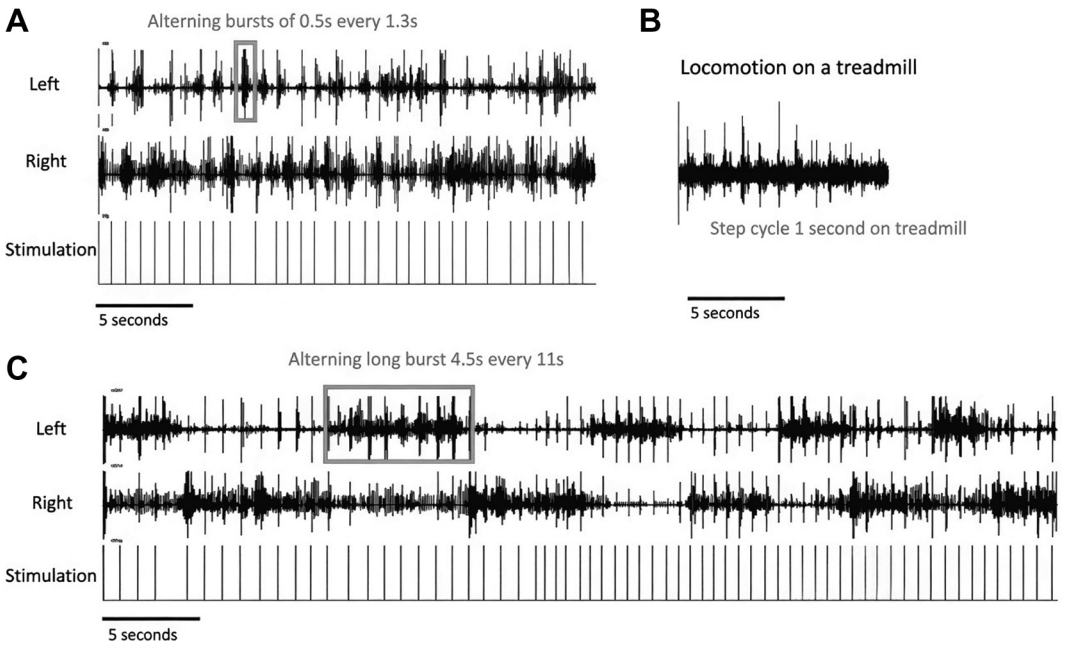



**Figure 4.** Repeated stimulation of the hindlimb representation of the pig's motor cortex causes alternating bursts in background activity, of variable durations. *A*: EMG trace showing rapid alternation in bursts of background activity in the tibialis anterior muscle with repeated stimulation of the hindlimb representation of the pig's motor cortex. *B*: EMG trace (vastus lateralis) showing locomotor cycles of the pig during treadmill walking at 0.3 m/s. *C*: EMG trace showing slow alternation in bursts of background activity in the tibialis anterior muscle with repeated stimulation of the hindlimb representation of the pig's motor cortex.



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intracortical or intravascular stimulation failed to elicit hindlimb movements (47, 48). Presumably, ungulates like the pig do not require much precise or complex control of hindlimb movements during natural behavior, rather relying on spinal central pattern generators and coarse movement modulation through brainstem circuits (49).

A key observation in our study is that repetitive ICMS of the hindlimb motor cortex elicited alternating bursts of hindlimb activity, coinciding with alternating hindlimb movements. This pattern of alternating activity resembles the hindlimb muscle activity elicited by repeated stimulation of the medullary reticular formation during fictive locomotion (50–52). In our study, stimulation was applied to the motor cortex rather than the reticular formation. Despite this, we hypothesize that stimulating the hindlimb representation of the pig motor cortex triggers hindlimb movements via the cortico-reticulo-spinal pathway. Unlike fictive locomotion studies such as Perreault et al. (50), we did not abolish phasic afferent feedback from the limbs. Consequently, we could not isolate the specific effects of repeated motor cortex stimulation on locomotor frequency as cleanly. Nevertheless, our hypothesis of a cortico-reticulo-spinal pathway for limb control in pigs is further supported by neuroanatomical tracing studies; pig's motor cortex neurons project to the contralateral cervico-thoracic anterior horn, ipsilateral mesencephalon, and bilateral medulla oblongata; no monosynaptic projection from the motor cortex to the lumbar region was ever described (42, 44). This hypothesis is also supported by the higher amplitude threshold required to activate the hindlimbs (300–500 µA) and the temporal summation needed to obtain clear responses. Forelimb responses were consistent from the first stimulation, at lower amplitude thresholds (<100 µA), more consistent with the ICMS literature. The reticular formation is a site of convergence of bilateral cortical inputs across species (53, 54). Taken together, these findings suggest that the porcine cerebral cortex can directly control forelimb movements through direct connections to cervical spinal neurons, but relies primarily on brainstem nuclei to modulate hindlimb activity.

This study has limitations. First, while the use of the micromanipulator was ideal for mapping the superior surface of the cortex, mapping the medial surface of the motor cortex, located in the interhemispheric fissure, proved more difficult. Because we preserved the midline dura to prevent hemorrhaging from the superior sagittal sinus, our direct visualization of the medial aspect was restricted. Consequently, we had to infer the functional representation of the medial motor cortex by analyzing the specific depths of our most medial electrode penetrations. Second, we measured flexor muscles in the hindlimbs (tibialis anterior), but signal from extensor muscles was too weak and artifacted to properly measure flexor-extensor alternation during alternated hindlimb movements. Third, the young age (2 mo) of the pigs used in the ICMS experiments may limit the generalization of our results to adult pigs, as the thickness of the motor cortex increases with age, accompanied by changes in cell density due to myelination and dendritic development (5). Despite these limitations, our study provides an updated characterization of the pig motor cortex and its descending pathways, informing the design and interpretation of translational neurotechnology studies.

## DATA AVAILABILITY

Data will be made available upon reasonable request.

## SUPPLEMENTAL MATERIAL

Supplemental Videos S1 and S2: https://doi.org/10.5281/zenodo.20085237.

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## GRANTS

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## DISCLOSURES

M.B. and M.M. are co-inventors on a patent (CA3146323) for a device enabling cortical stimulation during movement. They are also shareholders in a start-up company developing invasive neurostimulation technologies based on this patent. None of the other authors has any conflicts of interest, financial or otherwise, to disclose.

## AUTHOR CONTRIBUTIONS

D.B., M.B., and M.M. conceived and designed research; D.B., H.D.-M., M.B., and M.M. performed experiments; D.B. analyzed data; D.B. and M.M. interpreted results of experiments; D.B. prepared figures; D.B. drafted manuscript; D.B., H.D.-M., M.B., and M.M. edited and revised manuscript; D.B., H.D.-M., M.B., and M.M. approved final version of manuscript.

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